impl Sample for ZExpFamily<2>

Michael Shoemate

This proof resides in **"contrib"** because it has not completed the vetting process.

Proves soundness of the implementation of `Sample` for `ZExpFamily<2>` in `mod.rs` at commit f5bb719 (outdated[1]).

1 Hoare Triple

Precondition

`self` represents a valid probability distribution.

Pseudocode

```
1 class ZExpFamily2: # analogous to impl Sample for ZExpFamily<1> in Rust
2     def sample(self, shift):
3         sample = shift + sample_discrete_gaussian(self.scale) #
4         if self.divisor is not None: #
5             sample %= self.divisor
6         return sample
```

Postcondition

Theorem 1.1. Either returns `Err(e)` independently of the input `shift`, or returns a sample from the additive mechanism defined by `self` shifted by `shift`. When `self.divisor` is `None`, this is `shift + Z`. When `self.divisor = m`, this is `(shift + Z) mod m`.

Proof. By the precondition, since `self` represents a valid probability distribution, then by the definition of `ZExpFamily`, `self.scale` is non-negative.

Since the preconditions for are met (non-negative scale), the sample drawn on line 3 has the base discrete-Gaussian law encoded by `self.scale`.

If `self.divisor` is absent, line 3 returns `shift + Z`, establishing the non-modular case. If `self.divisor` is present, line 4 deterministically reduces the result modulo the divisor, which is exactly the wrapped additive mechanism encoded by `self`. Therefore the implementation satisfies the postcondition in either branch. □

[1] See new changes with `git diff f5bb719..40879b13 rust/src/measurements/noise/mod.rs`